

Effective Archimedean Stark Theorems over Real Quadratic Fields:

Quadratic Support, Shintani Transfer, and Higher-Order Packets

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Abstract

We prove a uniform formula for every one-place real-quadratic Stark invariant whose differenced Fourier support is quadratic. The formula is an explicit product of relative units, with rational exponents computed from class numbers, imprimitive Euler factors, and an exact relative-regulator index. Beyond this quadratic floor, we translate Shintani's index-two theorem into the one-place notation, give a general all-embeddings height-rigidity criterion, and use them to identify seven explicit packets with support orders up to ten. The examples include two order-six packets, an order-six packet at a ramified 3-power conductor, and an order-ten packet. Two structural lemmas prove that an absolutely abelian one-place ray field has trivial differenced support and that nontrivial support forces even Shintani index. The latter is independently consistent with all 446 genuinely reconstructed odd-index cases in a separate census. Exact and Arb certificates accompany the written proofs; no unproved Stark conjecture or floating-point recognition enters a proof.

1 Introduction and claim boundary

Let K be real quadratic and let $\mathfrak{m} = \mathfrak{f}\infty_2$ be a ray modulus containing one real place. For $K = \mathbb{Q}(\sqrt{D})$ with $D > 0$, our convention is that ∞_2 is the negative-square-root embedding $\sqrt{D} \mapsto -\sqrt{D}$, which is the first real place in the pinned PARI ordering; ∞_1 is the positive-square-root embedding. For $A \in \text{Cl}_{\mathfrak{m}}(K)$, let R be the sign class and put

$$Z_{\mathfrak{m}}(s, A) = \zeta_{\mathfrak{m}}(s, A) - \zeta_{\mathfrak{m}}(s, RA), \quad X_A = \exp Z'_{\mathfrak{m}}(0, A). \quad (1)$$

The numbers X_A are positive real analytic invariants. We call the finite Artin-indexed tuple

$$\mathcal{X}_{\mathfrak{m}} = (X_A)_{A \in \text{Cl}_{\mathfrak{m}}(K)}$$

the *one-place packet*; repeated values may be suppressed when a smaller Artin orbit is displayed. The word “packet” carries no algebraicity assumption. Stark's rank-one conjecture predicts algebraic units with the prescribed Artin action, but that conjecture is not known in this generality. The purpose of this paper is to identify explicit packets only where a proved theorem already supplies algebraicity and where the remaining analytic-to-algebraic comparison can be certified.

The word *unconditional* has this precise meaning throughout: each displayed identity follows from a published proved theorem under hypotheses checked exactly in the case bundle. It does not mean that the general real-quadratic rank-one Stark conjecture is proved. Shintani's theorem is used only in its index-two range [1], and the quadratic cases use the analytic class-number formula. Numerical PARI recognition, when retained as a cross-check, is outside the proof chain.

Terminology and quantitative conventions

Fourier inversion of (1) involves precisely the ray characters χ for which $\chi(R) = -1$. The *support order* is the maximum order of such a character. Labels such as RQ-000190 are stable locators in the accompanying corpus; they are not mathematical notation, and every use below also states the field and modulus.

For Engine B, a *safe exponent* is an explicitly computed integer m for which Shintani’s theorem proves that every X_A^m is algebraic in the asserted ray field. If the analytic and algebraic logarithms differ by balls of radius at most δ , the height comparison bounds the quotient by $m\delta$. The reported *margin* is

$$\mathcal{M} = \frac{V_{\min}}{m\delta}, \quad (2)$$

where $V_{\min}$ is the minimum applicable Voutier lower bound over the certified degree window. Thus $\mathcal{M} > 100$ is the pre-registered acceptance condition. A *cone evaluator* is the finite Shintani–Yamamoto decomposition of a partial zeta derivative into double-sine or Mellin-integral terms; all such terms below are evaluated as Arb balls.

How the results fit together

There are two theorem mechanisms. Engine A is a uniform closed-form formula for quadratic Fourier support. Engine B reaches higher character orders when Shintani’s maximal absolutely abelian subfield has index two: Proposition 6 translates the algebraicity theorem and Lemma 7 turns certified archimedean comparisons into exact identities. The order-six cases on either side of ramification isolate the index/wildness condition, rather than character order, as the relevant obstruction. Lemmas 8 and 9 explain two further structural boundaries.

Main theorems

Theorem 1 (Uniform quadratic-support theorem). *For every real quadratic K , every one-place modulus $\mathfrak{m} = \mathfrak{f}\infty_2$, and every packet whose differenced Fourier support consists only of quadratic characters, each X_A is an explicit rational power of a product of relative units. Corollary 4 gives one closed product formula whose exponents are expressed entirely in terms of exact class numbers, roots of unity, imprimitive Euler factors, and relative-regulator indices. No numerical approximation is required to state or prove the formula.*

Theorem 2 (Selected higher-order packets). *Use the integral basis $1, \omega$, where $\omega = (1 + \sqrt{D})/2$ for $D \equiv 1 \pmod{4}$ and $\omega = \sqrt{D}$ otherwise, and let the displayed HNF matrix denote the finite ideal $\mathfrak{f}$ in that basis. For every row below, the packet $\mathcal{X}_{\mathfrak{f}\infty_2}$ is the Artin-labeled set of positive roots of the relative polynomial printed in Section 4.*

case	K	HNF($\mathfrak{f}$)	support	m	margin
RQ-000190	$\mathbb{Q}(\sqrt{7})$	$\begin{bmatrix} 7 & 0 \\ 0 & 1 \end{bmatrix}$	2, 6	4032	> 5688
RQ-000419	$\mathbb{Q}(\sqrt{14})$	$\begin{bmatrix} 7 & 0 \\ 0 & 1 \end{bmatrix}$	2, 6	4032	> 7315
RQ-000108	$\mathbb{Q}(\sqrt{5})$	$\begin{bmatrix} 15 & 6 \\ 0 & 3 \end{bmatrix}$	4	2880	> 2460
RQ-000021	$\mathbb{Q}(\sqrt{2})$	$\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$	2, 6	2016	> 4261
RQ-002057	$\mathbb{Q}(\sqrt{57})$	$\begin{bmatrix} 9 & 3 \\ 0 & 3 \end{bmatrix}$	2, 6	2592	> 748
RQ-002955	$\mathbb{Q}(\sqrt{77})$	$\begin{bmatrix} 7 & 3 \\ 0 & 1 \end{bmatrix}$	2, 6	4032	> 5151
RQ-001107	$\mathbb{Q}(\sqrt{33})$	$\begin{bmatrix} 11 & 5 \\ 0 & 1 \end{bmatrix}$	2, 10	15840	> 5817

Here m is the safe exponent of Proposition 6, and “margin” has the meaning in (2). Each row includes exact ray-field identification, unconditional class and unit groups, Sturm isolation, all-embeddings Arb bounds, and exact Artin labels.

Historical scope. A dated literature search found no earlier unconditional oriented examples with support order six or ten, so these are apparently the first examples in those order classes. This historical observation is not part of either theorem and is not asserted as a universal priority claim.¹

Theorem 2 is deliberately a selected-results theorem, not a census-completeness statement. Counts, conductor trends, and a complete frontier taxonomy require the separately audited census and will appear elsewhere. This separation prevents provisional classification data from entering a theorem proof. The prior numerical and conditional computations of Dummit–Sands–Tangedal and Cohen–Roblot remain prior work rather than claims displaced here [6, 7, 8, 9]. More precisely, the 1997 Dummit–Sands–Tangedal computations concern totally real cubic bases and are numerical; their 2003 exponent-two theorems overlap the theorem class of Engine A, which is consequently not advertised as a historical first. Cohen–Roblot construct Hilbert class fields conditionally from Stark units and verify the resulting fields independently, whereas every promoted object here is a one-place ray field with nontrivial finite modulus. Thus these are mathematically adjacent precedents, not identical class-field objects. The order-six and order-ten historical statements refer only to the dated search described above.

2 The quadratic-support theorem

Let $G = \text{Cl}_m(K)$, and suppose every character χ with $\chi(R) = -1$ has order two. Write χ° for the associated primitive Hecke character and L_χ/K for the quadratic extension it cuts out. The selected real place becomes complex in L_χ , while the other real place splits, so L_χ has signature $(2, 1)$.

Let ϵ_K generate $E_K/\mu(K)$. Modulo torsion, use the logarithmic embedding

$$\lambda_L(u) = (\log |\tau_1 u|, \log |\tau_2 u|, 2 \log |\tau_3 u|), \quad (3)$$

where τ_1, τ_2 are real and τ_3 is one of the complex embeddings. Let u_χ generate the primitive rank-one kernel of

$$N_{L_\chi/K} : E_{L_\chi}/\mu(L_\chi) \longrightarrow E_K/\mu(K),$$

and orient it by $\log |\tau_1 u_\chi| > 0$. In any exact basis of $E_{L_\chi}/\mu(L_\chi)$, write v_K and v_χ for the coordinate columns of ϵ_K and u_χ , and define

$$I_\chi = [E_{L_\chi}/\mu(L_\chi) : \langle E_K/\mu(K), u_\chi \rangle] = |\det(v_K, v_\chi)|. \quad (4)$$

Proposition 3 (Explicit quadratic formula). *With E_χ the exact product of imprimitive Euler factors at zero,*

$$L'_m(0, \chi) = E_\chi \frac{h_{L_\chi}}{h_K} \frac{w_K}{w_{L_\chi}} \frac{2}{I_\chi} \log |\tau_1 u_\chi|. \quad (5)$$

Moreover

$$Z'_m(0, A) = \frac{2}{|G|} \sum_{\chi(R)=-1} \chi(A)^{-1} L'_m(0, \chi). \quad (6)$$

Consequently a common denominator in (5)–(6) gives an explicit power identity for every X_A .

Proof. For a primitive quadratic χ° , $\zeta_{L_\chi}(s) = \zeta_K(s) L(s, \chi^\circ)$. The analytic class-number formula at zero, with the same regulator normalization as (3), is

$$\zeta_F(s) = -\frac{h_F R_F}{w_F} s^{r_1(F)+r_2(F)-1} + O\left(s^{r_1(F)+r_2(F)}\right).$$

¹The search record names the sources, versions, hashes, and date. Kopp's arXiv:2411.06763 was checked again on 30 July 2026; the current version remains v3, dated 3 May 2025, and its algebraicity applications are conditional.

Since the unit ranks of K and L_χ are one and two, respectively, division gives

$$L'(0, \chi^\circ) = \frac{h_{L_\chi}}{h_K} \frac{w_K}{w_{L_\chi}} \frac{R_{L_\chi}}{R_K}. \quad (7)$$

We next prove the regulator identity without suppressing the mixed signature. Put $a = \log |\tau_1 \epsilon_K|$, where τ_1, τ_2 lie over the split real place of K . The product formula gives

$$\lambda_L(\epsilon_K) = (a, a, -2a).$$

For the relative unit u_χ , exact norm one modulo torsion gives

$$\lambda_L(u_\chi) = (b, -b, 0), \quad b = \log |\tau_1 u_\chi| > 0.$$

The determinant of the first two coordinates is $2|a|b$. Since $R_K = |a|$, the regulator of $\langle E_K, u_\chi \rangle$ is $2R_K \log |\tau_1 u_\chi|$. The same subgroup has index I_χ in the full free unit group, so its regulator is $I_\chi R_{L_\chi}$. Therefore

$$\frac{R_{L_\chi}}{R_K} = \frac{2}{I_\chi} \log |\tau_1 u_\chi|. \quad (8)$$

If $\mathfrak{m}/\mathfrak{f}_{\chi^\circ}$ is the imprimitive part, then

$$L_{\mathfrak{m}}(s, \chi) = L(s, \chi^\circ) \prod_{\mathfrak{p}|\mathfrak{m}/\mathfrak{f}_{\chi^\circ}} (1 - \chi^\circ(\mathfrak{p}) N\mathfrak{p}^{-s}).$$

At $s = 0$ the product is the exact integer E_χ . If one factor vanishes, both this product and $L(s, \chi^\circ)$ vanish at zero, so $L'_{\mathfrak{m}}(0, \chi) = 0$; otherwise combining (7) and (8) proves (5).

Finally, character inversion gives

$$\zeta_{\mathfrak{m}}(s, A) = \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \chi(A)^{-1} L_{\mathfrak{m}}(s, \chi).$$

Subtracting the formula with A replaced by RA multiplies the χ -term by $1 - \chi(R)^{-1}$, which is two precisely for $\chi(R) = -1$ and zero otherwise. Differentiation proves (6). Every surviving character is quadratic, hence every coefficient is rational. If q clears their denominators, exponentiation yields

$$X_A^q = \prod_{\chi(R)=-1} |\tau_1 u_\chi|^{n_{A,\chi}} \quad (n_{A,\chi} \in \mathbb{Z}), \quad (9)$$

after the exact factors in (5) have been absorbed into the integers $n_{A,\chi}$. This proves Proposition 3 and Theorem 1. $\square$

Corollary 4 (Closed packet formula). *Under the hypotheses of Theorem 1, put*

$$c_{A,\chi} = \frac{4}{|G|} \chi(A)^{-1} E_\chi \frac{h_{L_\chi}}{h_K} \frac{w_K}{w_{L_\chi}} \frac{1}{I_\chi}.$$

Then the positive real invariant at the selected embedding is

$$X_A = \prod_{\chi(R)=-1} |\tau_1 u_\chi|^{c_{A,\chi}}.$$

Every exponent $c_{A,\chi}$ is rational and is computed by exact finite arithmetic.

Remark 5. *The word “uniform” refers to this single character product, not to one field-independent named unit. For fixed $(K, \mathfrak{m}, A)$, the supported characters form a finite set, and substituting their exact class numbers, Euler factors, unit kernels, and determinant indices reduces the corollary immediately to a concrete product of explicitly computed relative units.*

The theorem is uniform; its proof does not enumerate the census. There are 1,560 nontrivial eligible occurrences in the frozen range, involving 2,232 supported character occurrences and 912 distinct quartic fields. These figures describe a verification queue, not a completeness claim here. Each application still certifies its primitive conductor, field, class numbers, Euler factors, exact norm map, primitive kernel, determinant, and orientation. The dimension-four and dimension-eight anchors independently replay (5), including relative index $I_\chi = 2$ in both dimension-eight cases.

3 Shintani transfer and height rigidity

Let $H = H_{\mathfrak{m}}$ be the one-place ray class field of $\mathfrak{m} = \mathfrak{f}_{\infty 2}$, and let $\mathcal{N}$ be its normal closure over $\mathbb{Q}$. Engine B first checks Shintani's conditions (0-3), (0-6), and (0-9), including

$$[H : H \cap \mathbb{Q}^{\text{ab}}] = 2. \quad (10)$$

The maximal absolutely abelian subfield is the fixed field of $[\text{Gal}(\mathcal{N}/\mathbb{Q}), \text{Gal}(\mathcal{N}/\mathbb{Q})]$. The imaginary quadratic field and transfer conductor below are then reconstructed a second time from their intrinsic ray class fields.

More precisely, an Engine-B transfer route consists of an imaginary quadratic subfield $k \subset \mathcal{N}$, a finite modulus $\mathfrak{c}$ of k , and a k -isomorphism

$$\iota : H_{\mathfrak{c}}(k) \xrightarrow{\sim} \mathcal{N}.$$

Here $\mathcal{N}/k$ is abelian, $\mathfrak{c} = \text{cond}(\mathcal{N}/k)$, and $H_{\mathfrak{c}}(k)$ denotes the corresponding ray class field. Thus “transfer conductor” below means this Artin conductor, not a conductor guessed from a discriminant. In every selected case it is computed by `rnfconductor` and the isomorphism is independently constructed by `bnrclassfield`; the two routes printed for a case must produce the same $\mathcal{N}$.

Proposition 6 (Shintani transfer in one-place notation). *Let $C = \text{Cl}_{\mathfrak{f}_{\infty 1} \infty_2}(K)$, and suppose exact fiber identities identify Shintani's invariant $X_{\mathfrak{f}}(\mathfrak{c}, G)$ with X_A in (1). Assume Shintani's conditions (0-3), (0-6), and (0-9), and let k and $\mathfrak{c}$ be respectively the imaginary quadratic field and finite transfer conductor occurring in his proof.*

For $\mathfrak{d} \mid \mathfrak{c}$, let

$$h(\mathfrak{d}) = |\text{Cl}_{\mathfrak{d}}(k)|, \quad w(\mathfrak{d}) = |\{\zeta \in \mu(k) : \zeta \equiv 1 \pmod{\mathfrak{d}}\}|,$$

put

$$n(\mathfrak{d}) = w(\mathfrak{d}) \frac{h(\mathfrak{c})}{h(\mathfrak{d})}, \quad (11)$$

and, for $\mathfrak{d} \neq \mathcal{O}_k$, let $f_{\mathfrak{d}}$ be the positive generator of $\mathfrak{d} \cap \mathbb{Z}$. If r clears the explicitly computed real-side distribution and induction denominators, then

$$\begin{aligned} m_{\mathcal{O}_k} &= 12h_k n(\mathcal{O}_k), \\ m_{\mathfrak{d}} &= 12f_{\mathfrak{d}} n(\mathfrak{d}) \quad (\mathfrak{d} \neq \mathcal{O}_k), \\ m &= \text{lcm}(r, (m_{\mathfrak{d}})_{\mathfrak{d} \mid \mathfrak{c}}) \end{aligned} \quad (12)$$

has the following property:

$$X_A^m \in \mathcal{O}_H^\times, \quad (X_A^m)^{\text{Art}(B)} = X_{AB}^m. \quad (13)$$

At every embedding of H above the nonsplit real place, $|X_A^m| = 1$.

Proof. Shintani's Theorem 1 identifies the logarithm of $X_{\mathfrak{f}}(c, G)$ with the full-ray zeta differences over the two fibers of A and RA ; the assumed exact fiber identity therefore gives X_A . Under (0-3), (0-6), and (0-9), Shintani's Theorem 2 supplies the index-two algebraicity construction [1].

In Shintani's equation (8), the distribution index is exactly (11). For $\mathfrak{d} \neq \mathcal{O}_k$, equations (4) and (9) contain $|\varphi_{\mathfrak{d}}|^{1/(6f_{\mathfrak{d}})}$ and then the power $1/n(\mathfrak{d})$. Replacing an absolute value by the product with its conjugate contributes a factor two, so $12f_{\mathfrak{d}}n(\mathfrak{d})$ clears the row. At the trivial divisor, equation (6) replaces $f_{\mathfrak{d}}$ by h_k . Proposition 4 is an integral product of ratios of these $W_{\mathfrak{c}}$ -invariants, so it introduces no further rational power. The separately computed integer r clears the real-side distribution or induction step. Raising to (12) therefore clears every exponent in the construction. Shintani's Proposition 5 gives the unit assertion, the Artin covariance, and modulus one at the nonsplit real embeddings, proving (13). $\square$

For the headline case $K = \mathbb{Q}(\sqrt{7})$, $\mathfrak{f} = \begin{bmatrix} 7 & 0 \\ 0 & 1 \end{bmatrix}$, the short transfer route has $k = \mathbb{Q}(i)$ and $\mathfrak{c} = (7)$. The complete divisor table is

$\mathfrak{d}$	$h(\mathfrak{d})$	$w(\mathfrak{d})$	$n(\mathfrak{d})$	$f_{\mathfrak{d}}$	$m_{\mathfrak{d}}$
$\mathcal{O}_k$	1	4	48	1	576
(7)	12	1	1	7	84

(14)

and the real-side indices are 6 and 1. Hence $\text{lcm}(576, 84, 6) = 4032$. This is a complete instance of Proposition 6; the longer $\mathbb{Q}(\sqrt{-7})$ reconstruction gives the same normal closure and safe power independently.

Lemma 7 (All-embeddings height rigidity). *Let H be a number field of degree D . Suppose $(X_A^m)_A \subset \mathcal{O}_H^\times$ is an algebraic family with Artin covariance, and let $(\alpha_A)_A \subset \mathcal{O}_H^\times$ be an exactly Artin-labeled positive candidate family. For each archimedean place v of H , suppose either*

$$|\log |X_A|_v - \log |\alpha_A|_v| \leq \epsilon_v \quad (15)$$

or both powered absolute values are proved to be one. Put

$$B_A = \frac{m}{D} \sum_{v|\infty} n_v \epsilon_v, \quad n_v = [H_v : \mathbb{R}], \quad (16)$$

with $\epsilon_v = 0$ in the second case. If

$$B_A < \min \left\{ \frac{1}{2} \log \frac{1 + \sqrt{5}}{2}, \min_{3 \leq d \leq D} \frac{1}{4d} \left(\frac{\log \log d}{\log d} \right)^3 \right\}, \quad (17)$$

then $X_A = \alpha_A$.

Proof. The quotient $\eta_A = X_A^m / \alpha_A^m$ is a unit of H . Its finite-place contribution to the Weil height is zero. Equations (15)–(16) and the definition of logarithmic height give

$$h(\eta_A) = \frac{1}{D} \sum_{v|\infty} n_v \max(0, \log |\eta_A|_v) \leq B_A.$$

If $d = [\mathbb{Q}(\eta_A) : \mathbb{Q}] \geq 3$ and η_A is not torsion, Voutier's explicit lower bound contradicts (17) [5]. A non-torsion quadratic unit has height at least $\frac{1}{2} \log((1 + \sqrt{5})/2)$, and a rational unit is ± 1 . Thus η_A is a root of unity. At the distinguished real embedding both X_A and α_A are positive, so $\eta_A = 1$; taking the unique positive m -th root proves the claim. $\square$

For every row of Theorem 2, the embeddings over the split real place are all represented by the printed packet roots and the cone enclosures. At the remaining archimedean places, Proposition 6 gives modulus one for X_A^m , and the reciprocal candidate polynomial is checked exactly, after the substitution $T = X + X^{-1}$, to have its remaining roots on the unit circle. Hence these places contribute $\epsilon_v = 0$ in (16).

The algebraic candidates are certified without a GRH assumption: each class and unit computation uses `bnfinit(P,1)`, followed by the full call `bnfcertify(bnf)` returning 1; the quotient-only flag is not used. Exact `rnfconductor` and `bnrclassfield` round trips identify the relative ray fields, Sturm sequences isolate the positive roots, and split-prime Frobenius actions label them.

Effective but not uniformly cheap. The required raw radius is $O(V_{\min}/m)$. Arb precision therefore grows with $\log m$, but conductor size, packet degree, and cone count also grow, and no uniform complexity claim is made. The order-ten case below has $m = 15840$ and remains comfortable; cases with much larger exponents can exceed the fixed per-case resource cap and belong to the frontier. This is a stated limit of Engine B, not evidence against its completed cases.

4 The order-six and order-ten packets

4.1 $\mathbb{Q}(\sqrt{7})$ at the ramified prime above seven

For $K = \mathbb{Q}(\sqrt{7})$ and $\mathfrak{m} = \mathfrak{p}_7 \infty_2$, the one-place and two-place ray groups are C_6 and $C_6 \times C_2$. The support orders are two and six and (10) holds. The commutator-fixed calculation produces $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-7})$, and their intrinsic conductor reconstructions give the same degree-24 normal closure. The $\mathbb{Q}(i)$ route has two divisor rows and safe exponent 4032.

The packet polynomial over K is

$$\begin{aligned} P_7(X) = & X^6 - (6 + 2\sqrt{7})X^5 + (22 + 8\sqrt{7})X^4 \\ & - (34 + 13\sqrt{7})X^3 + (22 + 8\sqrt{7})X^2 \\ & - (6 + 2\sqrt{7})X + 1. \end{aligned} \tag{19}$$

It is irreducible over K , defines the one-place ray field, has six positive Sturm-isolated roots, and is Artin-labeled by the split primes 19 and 31.

For the class represented by $(3)^r$, an exact Yamamoto domain has generators

$$\lambda_r = (7 - 2\sqrt{7})/3^r, \quad \lambda_r(8 + 3\sqrt{7}).$$

Its determinant is nine and its half-open parallelotope contains exactly nine lattice points. The resulting 27 independent double-sine evaluations are enclosed without a PARI L -value. At exponent 4032, the powered height upper bound is 9.1903×10^{-9} . The minimum Voutier lower bound in degrees three through 24 is 5.2279×10^{-5} , so the margin exceeds 5688. The degree-one and degree-two fallbacks finish the proof of (19).

This is the headline order-six instance. The same character order is the difficult component at the separate $\mathbb{Q}(\sqrt{21})$ wall, but here the index-two hypothesis holds and the packet closes. Thus the character order itself is not the obstruction.

4.2 Replications and the ramified-prime-3 control

For $\mathbb{Q}(\sqrt{14})$ at $\mathfrak{p}_7$, both $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-2})$ reconstruct the degree-24 normal closure. The latter route has clearing exponents 576 and 84, with least common multiple 4032. The packet

is

$$\begin{aligned}
P_{14}(X) = & X^6 - (13 + 4\sqrt{14})X^5 + (85 + 22\sqrt{14})X^4 \\
& - (139 + 38\sqrt{14})X^3 + (85 + 22\sqrt{14})X^2 \\
& - (13 + 4\sqrt{14})X + 1.
\end{aligned} \tag{20}$$

Split primes 11 and 103 label the Artin generators. The 20-point cone evaluator gives powered height at most 7.1466×10^{-9} , hence margin greater than 7315.

The strongest boundary control is RQ-002057 over $\mathbb{Q}(\sqrt{57})$, with modulus of norm 27. The conductor is a power of the ramified prime 3 and the support still has order six. The bases $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-19})$ independently give the same degree-24 closure. On the shorter route the divisor exponents are 864, 324, 108, hence $m = 2592$. With $y = (1 + \sqrt{57})/2$, so $y^2 - y - 14 = 0$, its packet is

$$\begin{aligned}
P_{57}(X) = & X^6 - (30 + 9y)X^5 + (582 + 177y)X^4 \\
& - (2529 + 772y)X^3 + (582 + 177y)X^2 \\
& - (30 + 9y)X + 1.
\end{aligned} \tag{20a}$$

The fundamental positive unit has order three modulo the finite modulus, so a ray-class domain is the union of three adjacent cones, with 240 exact affine points per class. The powered height is at most 6.982×10^{-8} , giving margin greater than 748. Thus order-six support and ramified-prime-3 structure are both reachable; neither alone explains the separate $\mathbb{Q}(\sqrt{21})$ boundary.

4.3 Three further Shintani closures

The three remaining rows of Theorem 2 use exactly the Engine-B proof above; we print their routes and algebraic candidates so that no theorem row is supported only by a corpus identifier.

For RQ-000108, $K = \mathbb{Q}(\sqrt{5})$ and $\mathfrak{N}\mathfrak{f} = 45$. The two imaginary bases are $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-15})$, and both reconstruct the same degree-16 normal closure. With $y = (1 + \sqrt{5})/2$, the relative packet polynomial is

$$X^4 - (1 + 6y)X^3 + (9 + 9y)X^2 - (1 + 6y)X + 1. \tag{21}$$

It is the certified one-place ray field over K . At $m = 2880$, the powered height is at most 2.12465×10^{-8} , so the Voutier margin exceeds 2460.

For RQ-000021, $K = \mathbb{Q}(\sqrt{2})$ and $\mathfrak{f} = (7)$. The bases $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-14})$ reconstruct the same degree-24 closure. The packet polynomial is

$$\begin{aligned}
& X^6 - (10 + 6\sqrt{2})X^5 + (58 + 40\sqrt{2})X^4 \\
& - (129 + 90\sqrt{2})X^3 + (58 + 40\sqrt{2})X^2 \\
& - (10 + 6\sqrt{2})X + 1.
\end{aligned} \tag{22}$$

The cone boundary meets a split-prime face; the canonical value is the average of the upper and lower half-open conventions. Both are enclosed independently. At $m = 2016$, the powered height is at most 1.22671×10^{-8} , giving margin greater than 4261.

For RQ-002955, $K = \mathbb{Q}(\sqrt{77})$ and $\mathfrak{f} = \mathfrak{p}_7$ has HNF $\begin{bmatrix} 7 & 3 \\ 0 & 1 \end{bmatrix}$. The bases $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-11})$ give the same degree-24 closure. Put $y = (1 + \sqrt{77})/2$, so $y^2 - y - 19 = 0$. The packet is

$$\begin{aligned}
& X^6 - (15 + 3y)X^5 + (107 + 26y)X^4 \\
& - (217 + 54y)X^3 + (107 + 26y)X^2 - (15 + 3y)X + 1.
\end{aligned} \tag{23}$$

At $m = 4032$, the powered height is at most 1.01485×10^{-8} , and the margin exceeds 5151. In all three cases the exact ray-field identification, Sturm isolation, Artin labels, and the degree-one and degree-two fallbacks are included in the cited case records.

4.4 Order ten

For RQ-001107 over $\mathbb{Q}(\sqrt{33})$ and $\mathfrak{p}_{11}$, put $y = (1 + \sqrt{33})/2 > 0$, so $y^2 - y - 8 = 0$. The packet is

$$\begin{aligned} P_{33}(X) = & X^{10} - (8 + 4y)X^9 + (74 + 30y)X^8 - (294 + 125y)X^7 \\ & + (669 + 281y)X^6 - (871 + 368y)X^5 + (669 + 281y)X^4 \\ & - (294 + 125y)X^3 + (74 + 30y)X^2 - (8 + 4y)X + 1. \end{aligned} \quad (24)$$

Its relative degree is ten, so the certified ray field H/K has absolute degree 20. The normal closure used to compute the Frobenius action has degree 40, but the comparison quotient $\eta_A = X_A^m/\alpha_A^m$ lies in H , since both powered terms do. Therefore Lemma 7 needs only $d \leq 20$. The archived computation conservatively evaluated every degree through 80; on either window the minimum Voutier bound is

$$5.22795332222684 \dots \times 10^{-5}.$$

At $m = 15840$, the factor-100 raw-error target is $3.30047558221391 \dots \times 10^{-11}$; the achieved bound is 5.6731×10^{-13} , so the certified margin exceeds 5817. The degree-one and degree-two fallbacks are recorded separately. This support order is the first one in the present proved inventory divisible by the prime 5; its historical priority status is the survey-bounded status stated after Theorem 2.

5 Structural boundary lemmas

Lemma 8 (Absolutely abelian ray fields). *Let H/K be the one-place ray class field of a real quadratic field. If $H/\mathbb{Q}$ is Galois and abelian, then the sign class $R \in \text{Gal}(H/K)$ is trivial and the differenced Fourier support in (1) is empty.*

Proof. If the one-place field $H/\mathbb{Q}$ were abelian, the complex conjugations above the two real places of K would be conjugate: any lift of the nontrivial element of $\text{Gal}(K/\mathbb{Q})$ interchanges the places. In an abelian group conjugate elements are equal. The inertia element at the omitted real place is trivial in the one-place ray field, whereas the inertia element at the retained place is R . Hence $R = 1$. It follows immediately that no character satisfies $\chi(R) = -1$, so the differenced Fourier support is empty. $\square$

Engine A is not an exception: Proposition 3 reduces individual quadratic characters through (5)–(6), but it does not assert that the entire mixed-signature one-place ray field is absolutely abelian.

Lemma 9 (Index parity). *Let H/K be a one-place ray field and let $N/\mathbb{Q}$ be its genuine normal closure. If the sign class R is nontrivial, then*

$$2 \mid [H : H \cap \mathbb{Q}^{\text{ab}}]. \quad (32)$$

Equivalently, an odd Shintani index can occur only when the Fourier support of (1) is empty.

Proof. Put $G = \text{Gal}(N/\mathbb{Q})$, $A = \text{Gal}(N/K)$, and $B = \text{Gal}(N/H)$. Let $c_1, c_2 \in A$ be the real-place inertia involutions. A lift of the nontrivial automorphism of $K/\mathbb{Q}$ interchanges them, so they are conjugate in G . Hence $c_1 c_2^{-1} \in [G, G]$. Since $G/A \simeq C_2$, one has $[G, G] \subseteq A$. Moreover B is normal in A , so $[G, G]B$ is a subgroup of A . In $A/B = \text{Gal}(H/K)$, the involution at the omitted place is trivial and the one at the retained place is R . Thus the image of $[G, G]$ in A/B contains R . Its order is the degree of H over its fixed field. That fixed field is

$$H \cap N^{[G, G]} = H \cap \mathbb{Q}^{\text{ab}},$$

because $N^{[G,G]}$ is the maximal abelian subextension of $N/\mathbb{Q}$. Hence the image has order $[H : H \cap \mathbb{Q}^{\text{ab}}]$, which is even if $R \neq 1$. Finally, characters of a finite abelian group separate a nonidentity element of order two, so $R = 1$ is equivalent to empty differenced support. $\square$

The use of the genuine normal closure in Lemma 9 is essential. Conjugation coordinates at an unstable modulus do not define $[G, G]$. As an independent consistency check, a pre-registered replay over all 8,200 genuine normal-closure reconstructions found 446 odd indices greater than one. Every one had trivial sign class and empty differenced Fourier support; there were no exceptions.

A Certificates and provenance

The exact field computations use PARI/GP 2.15.4; analytic certificates use Python 3.12.3 and python-flint 0.9.0. The principal replays were run under Linux/x86-64 on an AMD EPYC 9354P virtual CPU. All analytic conclusions are Arb balls, never point estimates. AI-assisted systems proposed code, text, and adversarial checks but were not mathematical authorities: every accepted computation was rerun by the pinned exact or Arb pipeline and reviewed by the author. The disclosure immediately before the bibliography gives the tools and purposes.

The principal machine-readable records are

result	record
first order six	data/q7-p7-case-v1.json
second order six	data/q14-p7-case-v1.json
RQ-000108	data/rq000108-case-v1.json
RQ-000021	data/rq000021-case-v1.json
ramified-prime-3 control	data/q57-norm27-case-v1.json
RQ-002955	data/rq002955-case-v1.json
order ten	data/q33-p11-order10-case-v1.json
all archimedean places	artifacts/engine-b-archimedean-place-audit-v1.json
quadratic theorem	data/engine-a-uniform-theorem-v1.json
index-parity theorem	artifacts/results-paper-index-parity-lemma-v1.json
parity audit	artifacts/results-paper-odd-index-parity-audit-v1.json
literature perimeter and Kopp	data/literature-perimeter-v1.json; artifacts/
watch	kopp-arxiv-watch-2026-07-30.json

Artin-label replay

For each cyclic ray group below, r denotes the power of the generator in the pinned PARI ray coordinates. The entry following r is a rational Sturm interval containing exactly one positive root of the absolute packet polynomial. The cone evaluation for class r , Shintani's Artin covariance (13), and the exact ray-field isomorphism identify that root with X_r .

case	r : isolating interval
RQ-000190	0 : (5.325, 5.326), 1 : (2.713, 2.714), 2 : (2.251, 2.252), 3 : (.187, .188), 4 : (.368, .369), 5 : (.444, .445)
RQ-000419	0 : (20.431, 20.432), 1 : (.189, .190), 2 : (1.090, 1.091), 3 : (.048, .049), 4 : (5.289, 5.290), 5 : (.916, .917)
RQ-000108	0 : (7.892, 7.893), 1 : (2.242, 2.243), 2 : (.126, .127), 3 : (.445, .446)
RQ-000021	0 : (6.109, 6.110), 1 : (7.496, 7.497), 2 : (4.351, 4.352), 3 : (.163, .164), 4 : (.133, .134), 5 : (.229, .230)
RQ-002057	0 : (5.709, 5.710), 1 : (.028, .029), 2 : (.035, .036), 3 : (.175, .176), 4 : (34.732, 34.733), 5 : (27.792, 27.793)
RQ-002955	0 : (18.232, 18.233), 1 : (2.180, 2.181), 2 : (8.620, 8.621), 3 : (.054, .055), 4 : (.458, .459), 5 : (.116, .117)
RQ-001107	0 : (8.645, 8.646), 1 : (.276, .277), 2 : (1.245, 1.246), 3 : (.571, .572), 4 : (.236, .237), 5 : (.115, .116), 6 : (3.611, 3.612), 7 : (.802, .803), 8 : (1.748, 1.749), 9 : (4.234, 4.235)

Each record carries hashes of its exact and analytic dependencies. The supplementary manifest also records the full software versions, commands, source hashes, and failed computations; those provenance details are not part of the mathematical argument.

Data availability. The companion archive will be deposited before submission. It will contain the source, exact and Arb certificates, the complete SHA-256 manifest, and one-command replay instructions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the author used OpenAI GPT-5.6 for mathematical brainstorming, symbolic-computation scaffolding, code review, and manuscript editing, and Anthropic Claude for independent adversarial checking of statements and certificates. The author reviewed and verified all outputs, made all mathematical and editorial decisions, and takes full responsibility for the content of the article.

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